

# DBMS – PRODUCT AND CATEGORY MANAGEMENT

## Project Title: ShopSphere – Online E-Commerce Database Management System

### 1. Database Design

The ShopSphere Online E-Commerce Database Management System uses Product and Category as important entities for managing products available in the e-commerce platform. The Category table stores category information, while the Product table stores product details such as product name, price, brand, stock quantity, and description.

#### 1.1 Category Table

| Attribute     | Data Type     | Constraint       |
|---------------|---------------|------------------|
| Category_ID   | NUMBER        | Primary Key      |
| Category_Name | VARCHAR2(100) | NOT NULL, UNIQUE |
| Description   | VARCHAR2(200) | —                |

#### 1.2 Product Table

| Attribute           | Data Type     | Constraint  |
|---------------------|---------------|-------------|
| Product_ID          | NUMBER        | Primary Key |
| Product_Name        | VARCHAR2(100) | NOT NULL    |
| Category_ID         | NUMBER        | Foreign Key |
| Price               | NUMBER(10,2)  | NOT NULL    |
| Brand               | VARCHAR2(100) | —           |
| Stock_Quantity      | NUMBER        | NOT NULL    |
| Product_Description | VARCHAR2(500) | —           |

### 2. Primary and Foreign Key Relationships

The relationship between Category and Product is defined using Category\_ID.

- Category.Category\_ID → Primary Key
- Product.Product\_ID → Primary Key
- Product.Category\_ID → Foreign Key referencing Category.Category\_ID
- Relationship: One-to-Many (1:M)
- One category can contain many products, while each product belongs to a category.

### 3. Store Product Details and Table Creation

The following SQL statements create the Category and Product tables and store product details such as name, category, price, brand, stock quantity, and description.

### 3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(100) NOT NULL UNIQUE,  
4     Description VARCHAR2(200)  
5 );
```

Table created.

### 3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID NUMBER,  
5     Price NUMBER(10,2) NOT NULL,  
6     Brand VARCHAR2(100),  
7     Stock_Quantity NUMBER NOT NULL,  
8     Product_Description VARCHAR2(500),  
9     FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
10 );
```

Table created.

## 4. Product Insertion, Updating and Deletion

### 4.1 Category Records

```
SQL> INSERT INTO Category VALUES
  2  (1, 'Electronics', 'Electronic products and accessories');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (2, 'Fashion', 'Clothing and fashion products');

1 row created.

SQL>
SQL> INSERT INTO Category VALUES
  2  (3, 'Home Appliances', 'Useful appliances for home');

1 row created.
```

## 4.2 Product Records

```
SQL> INSERT INTO Product VALUES
  2  (101, 'Wireless Headphones', 1, 1499.00, 'Boat', 25,
  3  'Bluetooth wireless headphones');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2  (102, 'Cotton T-Shirt', 2, 599.00, 'Max', 40,
  3  'Comfortable cotton T-shirt');

1 row created.

SQL>
SQL> INSERT INTO Product VALUES
  2  (103, 'Electric Kettle', 3, 1299.00, 'Prestige', 15,
  3  'Stainless steel electric kettle');

1 row created.
```

## 4.3 Product Update

Update the price and stock quantity of Wireless Headphones.

```
SQL> UPDATE Product
  2  SET Price = 1399.00,
  3      Stock_Quantity = 30
  4  WHERE Product_Name = 'Wireless Headphones';

1 row updated.
```

#### 4.4 Product Deletion

Delete the product Cotton T-Shirt from the Product table.

```
SQL> DELETE FROM Product
  2  WHERE Product_Name = 'Cotton T-Shirt';

1 row deleted.
```

### 5. Category-wise Product Reports

Category-wise reports are generated to display products according to their categories and to count the number of products available in each category.

#### 5.1 Product Listing by Category

This report displays the category name along with the products belonging to each category.

```

CATEGORY_NAME
-----
PRODUCT_NAME
-----
    PRICE
-----
BRAND
-----
STOCK_QUANTITY
-----
Electronics
Wireless Headphones
    1399

CATEGORY_NAME
-----
PRODUCT_NAME
-----
    PRICE
-----
BRAND
-----
STOCK_QUANTITY
-----
Boat
    30

CATEGORY_NAME
-----
PRODUCT_NAME
-----
    PRICE
-----
BRAND
-----
STOCK_QUANTITY
-----

```

```

BRAND
-----
STOCK_QUANTITY
-----
Home Appliances
Electric Kettle
    1299

CATEGORY_NAME
-----
PRODUCT_NAME
-----
    PRICE
-----
BRAND
-----
STOCK_QUANTITY
-----
Prestige
    15

```

## 5.2 Products in Electronics Category

```

PRODUCT_NAME
-----
      PRICE
-----
BRAND
-----
STOCK_QUANTITY
-----
Wireless Headphones
      1399
Boat
      30

```

### 5.3 Product Count by Category

```

PRODUCT_NAME
-----
      PRICE
-----
BRAND
-----
STOCK_QUANTITY
-----
Wireless Headphones
      1399
Boat
      30

```

### Conclusion:

The ShopSphere Online E-Commerce Database Management System demonstrates database design using Product and Category entities. The task establishes primary and foreign key relationships, stores product information, performs insertion, updating, and deletion operations, and generates category-wise product reports. This design supports organized and efficient management of product and category information in the ShopSphere e-commerce database.